

Qiskit v2.x Practice Quiz

Questions

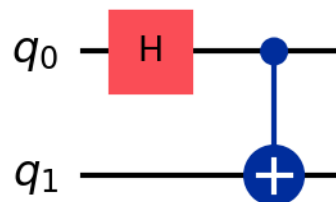
1. Which of these are OpenQASM 3 features? *(Select all that apply)*
 - A. while loops
 - B. angle
 - C. try/catch
 - D. barrier
2. If the observable is identity I, what expectation value should Estimator ideally return for any normalized state?
 - A. 0
 - B. -1
 - C. Depends on the number of shots
 - D. 1
3. What is the result of applying Z to $|1\rangle$?
 - A. $|0\rangle$
 - B. $|1\rangle$
 - C. $-|1\rangle$
 - D. $|+\rangle$
4. When is Sampler more appropriate than Estimator?
 - A. When you need bitstring measurement outcomes
 - B. When you need expectation values of observables
 - C. When you need to draw a circuit
 - D. When you need to create a backend
5. Which gate maps $|0\rangle$ to $|+\rangle$?
 - A. X
 - B. Z
 - C. H
 - D. CX
6. What is the purpose of transpilation?

- A. To convert a circuit into backend-compatible instructions
- B. To measure all qubits automatically
- C. To create a statevector
- D. To define an observable

7. Which state lies on the positive X axis of the Bloch sphere?

- A. $|0\rangle$
- B. $|1\rangle$
- C. $|+\rangle$
- D. $|-\rangle$

8. What state does the circuit below prepare?



- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

9. What does an X gate do to $|0\rangle$?

- A. Creates $|+\rangle$
- B. Maps it to $|1\rangle$
- C. Measures the qubit
- D. Adds a global phase only

10. What is the default value for NAME_OF_CREG in the sampler PUB output?

- A. clbit
- B. meas
- C. evs
- D. pubs

11. Which one of the following images is the output from the code below:

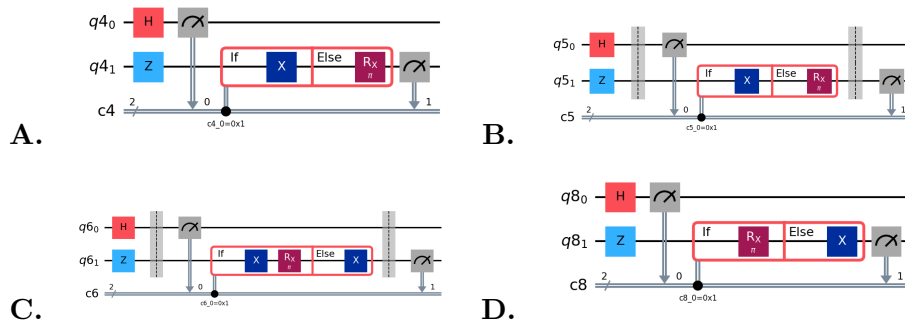
```

from qiskit import *
from math import pi

qubits = QuantumRegister(2)
clbits = ClassicalRegister(2)
qc = QuantumCircuit(qubits, clbits)
(q0, q1) = qubits
(c1, c2) = clbits

qc.h(0)
qc.z(1)
qc.barrier()
qc.measure(q0, c1)
with qc.if_test((c1,1)) as else_:
    qc.x(q1)
with else_:
    qc.rx(pi, q1)
qc.barrier()
qc.measure(q1, c2)
qc.draw(output="mpl", plot_barriers=False)

```



12. Which object is commonly used to represent Pauli observables in Qiskit?

- A. QuantumCircuit
- B. SparsePauliOp
- C. ClassicalRegister
- D. PassManager

13. What does the following circuit prepare?

```

qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)

```

- A. $|00\rangle$
- B. $|11\rangle$
- C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

14. Which OpenQASM 3 variable declaration corresponds to Qiskit's `ClassicalRegister` declaration?
- A. `bit`
 - B. `clreq[ ]`
 - C. `bit[ ]`
 - D. `clbit[ ]`
15. Why may SWAP gates be inserted during transpilation?
- A. To reset qubits
 - B. To satisfy hardware connectivity constraints
 - C. To increase shot count
 - D. To remove all noise
16. What does the `Sampler` primitive primarily return?
- A. Expectation values
 - B. Measurement samples or bitstring results
 - C. Only statevectors
 - D. Only transpiled circuits
17. How long can you access previously submitted jobs?
- A. Jobs are retained for 30 days after completion
 - B. Jobs are retained for 1 year after completion
 - C. Undeleted job data is retained for three years while you continue using the service
 - D. Jobs are permanently accessible unless manually deleted
18. Which code correctly creates a two-qubit quantum circuit?
- A. `QuantumCircuit()`
 - B. `QuantumCircuit(2)`
 - C. `Circuit(2)`
 - D. `QuantumRegister.create(2)`
19. What does layout refer to in transpilation?
- A. Mapping virtual qubits to physical qubits
 - B. Choosing the number of shots
 - C. Drawing the circuit
 - D. Choosing classical register names
20. What does the Bloch sphere represent?
- A. Only classical bits
 - B. Single-qubit pure states
 - C. Only two-qubit entangled states
 - D. Classical probability distributions only

Answer Key

1. S8-005: A, B, D
2. S6-009: D
3. S1-003: C
4. S5-002: A
5. S1-002: C
6. S4-001: A
7. S2-002: C
8. S3-004: C
9. S1-001: B
10. S7-003: B
11. S3-003: A
12. S6-002: B
13. S3-002: C
14. S8-003: C
15. S4-002: B
16. S5-001: B
17. S7-006: C
18. S3-001: B
19. S4-003: A
20. S2-001: B